



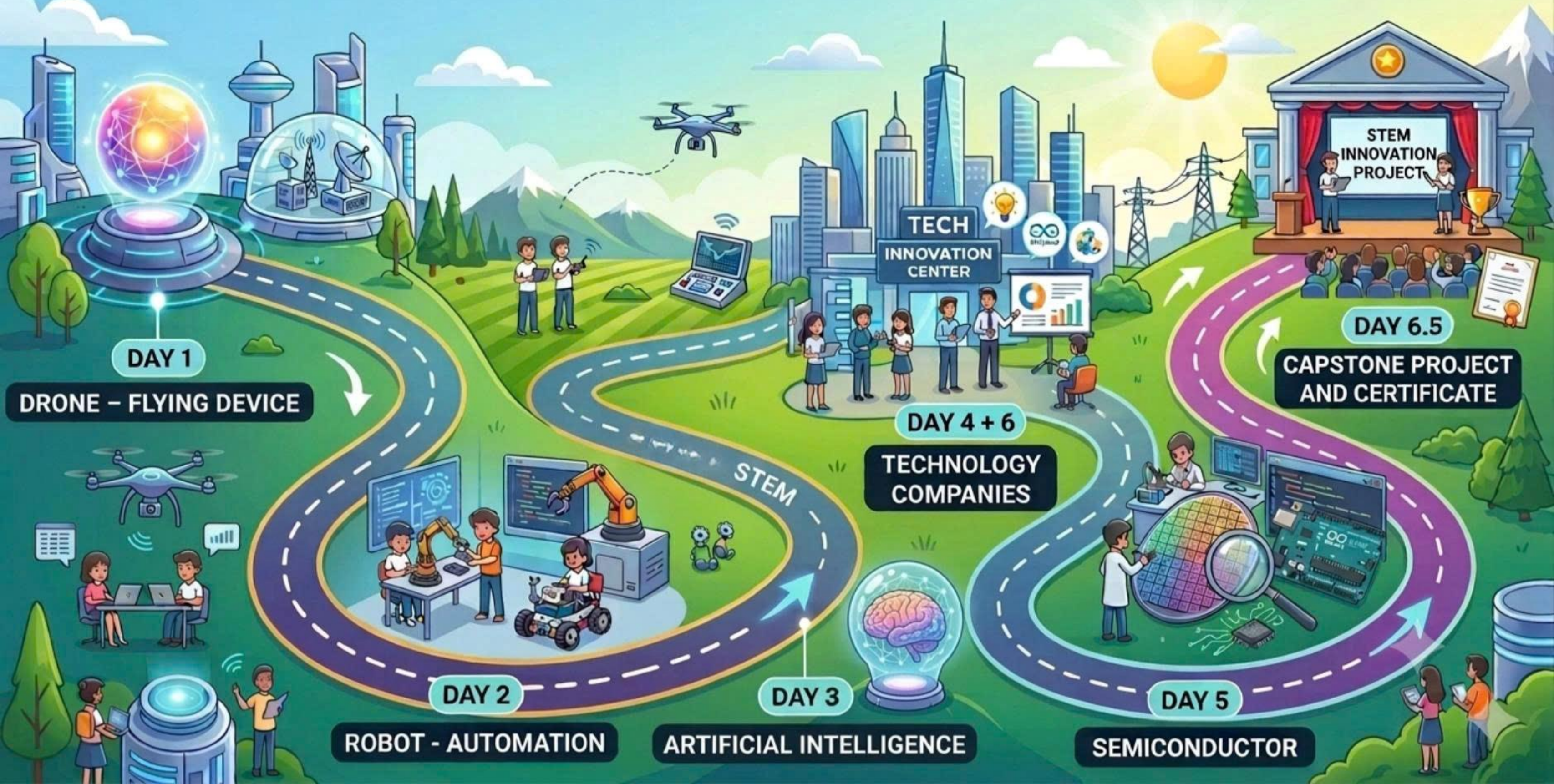
YOUR GROWTH IS OUR PRIDE

PROGRAM

TECH EXPERIENCE STEM



STEM COURSE ROADMAP – A JOURNEY OF TECHNOLOGICAL DISCOVERY



STEM CHALLENGE

I DRAW THE FUTURE OF SMART SOCIETY

TEAM - 5 MEMBERS



CONTEXT

In 2045, the world faces many challenges such as pollution, congestion, energy shortage, aging population... Your team's mission is to design a technology solution to help build a 'SMART SOCIETY'.



5 ROLES - 5 STRENGTHS

1. CEO STRATEGIST



- + Coordinate group
- + Define vision
- + Make the final decision

2. SOCIAL RESEARCHER



- + Identify the biggest problem
- + Gather ideas
- + Understand people's needs

3. TECH ENGINEER



- + Technology proposals
- + AI, IoT, Robot, Drone, Energy...
- + Engineering thinking

4. UX DESIGNER



- + Conceptual drawing
- + Product design, interface
- + User experience

5. INVESTOR



- + Feasibility assessment
- + Cost-benefit analysis
- + Business model
- + Who will pay?

1 PROBLEM IDENTIFICATION (10 min)

Each member writes 3 social problems on sticky notes.



2 GROUP IDEAS (10 min)

Categorize problems by topic groups.



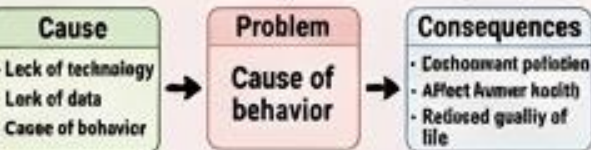
3 VOTING & SELECTION (5 min)

Each member has 3 votes. Which issue is most important?



4 SYSTEMS THINKING

Draw a cause-effect (problem-consequence) map.



7 DRAWING THE FUTURE OF SMART SOCIETY



5 STARTUP DESIGN

- Startup name: **GreenBrain AI**
- Slogan: **AI for a green city**
- Customers: **Cities, schools, businesses, citizens**
- Solution: **Autonomous AI waste-sorting robots combined with a smart data platform**
- Technology: **AI Vision, Robot, IoT, Cloud, Big Data, Renewable Energy**

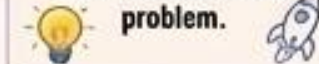


User App
Rewards - Redemptions
Build green cities

8 GENERATING NEW KNOWLEDGE

We discovered that...

Waste is not just a technology problem, but a data and behavior problem.



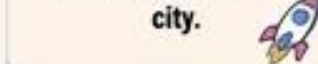
We learned that...

AI is only effective when combined with human behavioral change.



Our group's new idea is...

Build a Robot + AI + Education ecosystem to create a zero-waste city.



6 DECISION MAKING - IDEA SCORING

CRITERIA	MAX SCORE	GROUP SCORE	TOTAL
Solve a major problem	10		
Creativity	10		
Feasibility	10		
Social impact	10		
Business potential	10		
TOTAL	50		

9 DEMO DAY

Present your group's top 3 inputs to the class.

1. Problem
2. Solution
3. Technology
4. Business Model
5. Smart Society Drawing
6. New Knowledge Discovered





<https://github.com/haitnshtp/shtp-stem-smart-society-eng-template>

Wayfinder	Lotus Eco AI	Debug	Opera VX
Nguyen Tuan Khai	Le Minh Kiet	Ho Tran Chau Thao	Hoang Lam Tai
Henry Herzog	Do Hong An	Nguyen Minh Quan	Vu Chan Hung
Le Minh Dat	Pham Nhuong Duy Anh	Tri Nam An	Nguyen Lam Quoc Hung
Lam Chi Khang	Do Le Thuc Linh	Nguyen Bao Tuong	Tran Phan Lam
Tran Ha Chi	Hoang Nguyen Phuc Nguyen	Vu Tran Tat Thang	Nguyen Huynh Ngoc Tam



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Thank You



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(84-28) 37 360 051